

Sayed Huzaifa Mumit

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Education

University of Science and Technology Chittagong (USTC), Bangladesh [Ongoing]

Major in B.S. (Hons.) in Biochemistry and Biotechnology

Supervisor: Dr. Md. Monirul Islam

Affiliation: The Bangladesh Bioscience Research Group (BBRG)

UG Research: Computational Biology & Biophysics

Minors in

Biophysical Chemistry Biostatistics Computer Science Pharmaceutical Chemistry Spectroscopy

Profile Summary

I am an undergraduate student in Biochemistry and Biotechnology with ongoing research involvement in RNA biology, genomics, immunology, and phytochemistry. Alongside my academic work, I have been independently exploring theoretical astrophysics since high school, focusing on gravitational dynamics, black hole behavior, and the relationship between spacetime geometry and matter. My goal is to build a solid mathematical foundation in classical mechanics, general relativity, and quantum theory so I can formalize these theoretical ideas into a testable framework. I have presented research in multiple national and international conferences and have experience publishing interdisciplinary projects in environmental sustainability and renewable energy. I am motivated by long-term curiosity, self-driven learning, and a commitment to connecting scientific understanding across fields.

Research Experiences

Bangladesh Bioscience Research Group (BBRG)

October 2024 – Present

Undergraduate Research & Primary Affiliation

RNA Biology, Genomics & CRISPR, Immunology, Computational Biology

Independent Research

2019 – Present

Theoretical Astrophysics (Quantum Gravity), Renewable Energy, Environmental Science

PHY Course	Focus
Quantum Mechanics	Operator formalism, Hilbert spaces, wave dynamics
General Relativity	Spacetime curvature and gravity as geometry
Differential Geometry	Manifolds, tensors, and curvature tools for GR
Fourier Analysis	Signal Processing and Time-Series Analysis

Academic Volunteering

Panelist, The Cambridge Panel: Academic

October 2025 – Present

Cambridge University Press — Participating in shaping academic publishing practices through feedback on research and author experience

Digital Footprint

[ResearchGate](#) | [Google Scholar](#) | [ORCID](#) | [LinkedIn](#) | [Github](#) | [Academia.edu](#) | [Web of Science](#)

Publications & Preprints

- 1* Methane Production via Sabatier Reaction Using Hydrogen from Rainwater Electrolysis and Captured Carbon Dioxide v1 (2023) | v2 (2025); DOI: [10.33774/coe-2025-lk4r7](https://doi.org/10.33774/coe-2025-lk4r7)
- 2* Eco-Friendly Water Purification Using Solar-Powered Distillation and Activated Carbon (2024) [*Expected Publication in 2026*]; Preprint DOI: [10.33774/coe-2024-ztkpb](https://doi.org/10.33774/coe-2024-ztkpb)
- 3* Sustainable Diapers and Pads from Water Hyacinth and Coconut Coir v1 (2024) | v2 (2025); DOI: [10.21203/rs.3.rs-5453339/v2](https://doi.org/10.21203/rs.3.rs-5453339/v2)
- 4* Investigation of the Effectiveness of Using Luffa Cylindrica to Reduce Water Consumption in Agriculture (2024); DOI: [10.62622/TEIEE.024.2.4.18-27](https://doi.org/10.62622/TEIEE.024.2.4.18-27)

Research Projects

I pursued these projects to sustain my engagement in research and refine experimental thinking, with an emphasis on learning and proof-of-concept rather than publication.

- 1* The Role of Buoyancy and Thermal Convection in Flame Orientation Under Gravitational Fields (2019-2020)
- 2* Luffa-Based Depth Filtration for Microplastic Removal Under Controlled Hydraulic Pressure (2024) [Finalist Project at XYLEM GLOBAL INNOVATION CHALLENGE 2024]
- 3* In silico Analysis of the Phytochemical Compounds from Plant *Abelmoschus* genus plant (2024-2025)
- 4* In silico Analysis of the Phytochemical Compounds from Plant *Colocasia* genus plant (2025)
- 45 In silico Analysis of the Phytochemical Compounds from Plant *Samanea* genus plant (2025)

Conference Papers (Presenting Author)

- 1* 6th International Conference on Biotechnology in Health and Agriculture — Global Network of Bangladeshi Biotechnologist. Prevalence of Iron Deficiency Anemia and Evaluation of Corrected Calcium as a Biochemical Parameter Among Young Adults in Chattogram, Bangladesh
- 2* RURS 1st International Students Conference 2024 — Trends in Microbial Culture and Antibiotic Sensitivity in Pediatrics
- 3* Xylem Global Water Challenge 2024, Stockholm, Sweden — Using Sponge Gourd for Purifying Microplastics
- 4* Stockholm Junior Water Prize 2023 — Methane Gas Production from Rainwater and CO₂
- 5* American Physical Society Physics Summit 2025, Anaheim, USA — Graviton and Anti-Graviton Charges on Space-Time Curvature and Gravitational Dynamics [Acceptance Only]

Peer Reviewer

International Peer Reviewer, International Journal of Plant and Soil Science (2025)

Manuscript: Optimizing Sponge Gourd Cultivation: Effects of Chemical Hybridizing Agents on Morphological Traits and Flower Development

[Web of Science Author Record](#)

Books

- 1* From Concept to Publication: A Comprehensive Guide to Preparing a High-Quality Research Manuscript for International Journals. [VISIT](#)
- 2* Meta Analysis of Skills: An All-Inclusive Manual. [VISIT](#)

Skills

Soft	Hard	Computational Tools
Leadership	Research	Python
Public Speaking	Phytochemical Analysis	Linux/UNIX Shell
Conflict Resolution	Molecular Docking	LaTeX
Adaptability	Network Pharmacology	Cytoscape, AutoDock, SwissADME

Languages

English	Bengali	Chittagonian	Arabic
Intermediate	Native	Native	Reading

Entrepreneurial & Leadership Experience

- 1* Co-Founder, A TimesM Global — Multi-sector startup (2023-2025)
- 2* Co-Founder, Manoshota (2021–2022) — Global mental health consulting, assisting 5,700 individuals in 13 hours

Scholarships & Fellowships / Grants

- 1* APS Diversity Forum Scholarship — American Physical Society (2024)
- 2* Wiki Scientist Summer Program Scholarship — Wikipedia (2024)
- 3* Oxbridge Scholarships — University of Oxford & Cambridge (2022 & 2023)
- 4* Academia Engineering Publication Grant (2024)
- 5* Trends in Ecological & Environmental Engineering Journal Grant (2024)
- 6* AB Memorial Fellowships (2022)

Awards & Recognitions (Selective)

- 1* Medalist, 16th National Bangladesh Astro-Olympiad (2022)
- 2* Finalist, Stockholm Junior Water Prize (2023)

- 3* Top-Ranked, People's Choice — Xylem Global Innovation Challenges (2024)
- 4* Jamboree Award, 10th Bangladesh & 3rd South Asian Association of National Scouts Jamboree (2019)
- 5* Runners Up, Essay Competition, District Level Chittagong, Bangladesh (2018)
- 6* Runners Up, Speech Competition, District Level Chittagong, Bangladesh (2018)

Professional Memberships

National

- 1* Graduate Member, BYLC Graduate Network (2021–Present)
- 2* Member, Astrophysics Division, American Physical Society (2024–Present)

International

- 2* Alumni Member, House of Volunteers Bangladesh (2023–Present)
- 2* Member, Biophysical Society (2025–Present)
- 2* Wiki Scientist, Wikimedia Foundation (2024–Present)
- 2* Member, American Society for Microbiology (2024–2025)
- 2* Singapore Global Network (2025–Present)

Special Academic Acceptances

- 1* Aspire Leaders Program, Harvard University (2024)
- 2* Wiki Scientist Summer Program, APS (2024)
- 3* Oxbridge Scholars Program, University of Oxford & Cambridge (2022 & 2023)

Teaching

- 1* APS Wiki Scientist, Wikipedia (2024–Present)
- 2* IAU OAO Venture Translator, International Astronomical Union (2024)
- 3* Academic Olympiad Trainer, Deviation Scientific Society (2023–Present)
- 4* Senior Patrol Leader, Chattogram Metropolitan Scouts (2018–2021)

Competitions & Research Challenges

- 1* International Contestant, MIT SOLVE (2024)
- 2* International Contestant, CERN Beamline for Schools (BL4S) (2024)
- 3* International Contestant, 3rd SAANSO & 10th Bangladesh Scouts Jamboree (2019)
- 4* Regional Round, Bangladesh Mathematical Olympiad
- 5* Regional Round, Bangladesh Physics Olympiad

References

Dr. Md. Monirul Islam

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Dr. Jeffrey M. Keefer

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Project Management & Information
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